

WRD SERIES (20 TO 2,000 CFM)

High-Efficiency Refrigerated Air Dryers

Non-Cycling Direct-Expansion Technology Delivering Constant +3°C Pressure Dew Point Under Extreme Tropical Ambient Conditions



+3°C PDP

ISO 8573-1 Class 4
Moisture Purity

20–2,000 CFM

Comprehensive Standard
Range

50°C MAX INLET

Tropical Condenser
Rating

ZERO AIR LOSS

Capacitive Condensate
Drain

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WRD Series refrigerated compressed air dryers remove moisture vapor from compressed air networks through precision thermodynamic chilling. Engineered specifically for Indian and GCC tropical environments with ambient temperatures reaching 45°C and air inlet temperatures up to 50°C, WRD dryers protect sensitive pneumatic machinery, CNC spindles, and automated paint lines from liquid water, emulsion sludge, and corrosion.

Working Principle: Saturated compressed air enters the high-efficiency 3-in-1 modular aluminum heat exchanger. In the air-to-air regenerator stage, incoming warm wet air is precooled by outgoing cold dry air, reducing refrigeration compressor power demand by up to 40%. Precooled air then passes into the direct-expansion evaporator where it is chilled to +3°C by eco-friendly R134a/R407C refrigerant. Moisture condenses into droplets and is separated by a multi-stage centrifugal demister before automated discharge through a zero-air-loss drain.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Air Compressor → Aftercooler (<45°C) → Wet Receiver Tank → Pre-Filter (3.0μ) → WRD Refrigerated Dryer (+3°C PDP) → Coalescing Filter (0.01μ) → Dry Receiver → Distribution Ring Main

⚡ 3-in-1 Monoblock Aluminum Heat Exchanger

Combines air-to-air precooler/reheater, direct-expansion evaporator, and high-velocity centrifugal water separator in a single corrosion-resistant brazed aluminum block. Eliminates interconnecting pipe friction to deliver ultra-low pressure drop under 0.2 bar.

❄️ Mechanical Hot Gas Bypass Modulation

Features pressure-balanced hot gas bypass injection that automatically modulates refrigerant flow from 0% to 100% pneumatic demand. Guarantees constant +3°C dew point without evaporator freezing, even during extended zero-demand standby.

🛡️ Copeland / Danfoss Heavy-Duty Compressors

Hermetic scroll and reciprocating refrigeration compressors engineered for continuous 24/7 industrial duty. Equipped with internal thermal motor protection, high/low pressure safety cutouts, and anti-vibration rubber isolation mounts.

💧 Zero Air Loss Electronic Capacitive Drain

Intelligent non-contact capacitive level sensor detects accumulated condensate and vents 100% liquid without escaping compressed air. Saves over ₹42,000 annually per 500 CFM installation compared to lossy timed solenoid drains.



CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: Nominal Working Pressure: 7.0 bar g (Max 16.0 bar g) | Max Inlet Temp: 50°C | Ambient Temp: 35°C | Pressure Dew Point: +3°C | Refrigerant: R134a / R407C

MODEL CODE	NOMINAL CFM	FLOW M ³ /HR	POWER KW	SUPPLY VOLTAGE	IN/OUT CONN	DIMENSIONS (L×W×H MM)	WEIGHT KG
WRD 20 S	20 CFM	34	0.45 kW	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550	38 kg
WRD 30 S	30 CFM	51	0.45 kW	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550	40 kg
WRD 40 S	40 CFM	68	0.45 kW	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550	42 kg
WRD 60 S	60 CFM	102	0.65 kW	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750	55 kg
WRD 80 S	80 CFM	136	0.65 kW	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750	58 kg
WRD 100 S	100 CFM	170	0.85 kW	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750	60 kg
WRD 125 S	125 CFM	212	0.85 kW	230V / 1Ph / 50Hz	1.5" BSP	500 × 550 × 750	65 kg
WRD 150 S	150 CFM	255	1.10 kW	230V / 1Ph / 50Hz	1.5" BSP	600 × 650 × 900	90 kg
WRD 200 S	200 CFM	340	1.25 kW	230V / 1Ph / 50Hz	1.5" BSP	600 × 650 × 900	95 kg
WRD 250 S	250 CFM	425	1.65 kW	230V / 1Ph / 50Hz	2" BSP	600 × 650 × 900	105 kg
WRD 300 S	300 CFM	510	1.65 kW	230V / 1Ph / 50Hz	2" BSP	600 × 650 × 900	110 kg
WRD 400 S	400 CFM	680	2.20 kW	230V / 1Ph / 50Hz	2.5" BSP	800 × 800 × 1100	180 kg
WRD 500 S	500 CFM	850	2.20 kW	230V / 1Ph / 50Hz	2.5" BSP	800 × 800 × 1100	190 kg
WRD 600 S	600 CFM	1,020	3.50 kW	415V / 3Ph / 50Hz	3" Flange	1200 × 850 × 1350	280 kg
WRD 750 S	750 CFM	1,275	3.50 kW	415V / 3Ph / 50Hz	3" Flange	1200 × 850 × 1350	300 kg
WRD 1000 S	1,000 CFM	1,700	5.50 kW	415V / 3Ph / 50Hz	4" Flange	1500 × 1000 × 1500	450 kg

ENGINEERING SELECTION & OPERATIONAL NOTES

- Correction Formula: Actual Capacity = Nominal CFM × Cp × Ct × Ca
- Operating pressure range: 4.0 to 16.0 bar g (High pressure models up to 40 bar available on request)
- Refrigerant R134a used for WRD 20–300; R407C used for WRD 400–2000.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

Textile Spinning & Weaving	Protects electronic yarn clearers, autoconers, and air-jet loom nozzles from moisture spots, rust jamming, and yarn contamination.
CNC Machine Tooling & Automotive	Eliminates moisture-induced solenoid sticking, spindle taper oxidation, and pneumatically clamped pallet positioning drift.
Automated Spray Painting	Prevents micro-blistering, cratering, and paint fish-eye defects in robotic automotive clear-coat booths.
Fiber Laser Cutting Systems	Supplies pure dry assist gas, shielding expensive collimator and focus optics from mist condensation and beam scattering.
PET Container Blow Molding	Pre-treats booster compressor air, eliminating condensate puddling inside blow molds and ensuring optical bottle clarity.

Engineering Benchmark: Win Equipments WRD vs. European Imports		Standard ISO 9001:2015 Quality
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark
Ambient Rating	Continuous 45°C (Peak 50°C) with oversized condenser	Derated above 35°C (Frequent trip-outs)
Heat Exchanger	Corrosion-resistant 3-in-1 brazed aluminum monoblock	Multi-joint copper tube-in-shell (High leak risk)
Condensate Drain	Integrated Zero-Air-Loss capacitive sensor (Standard)	Lossy mechanical float or optional timed solenoid
Service & Spares	Immediate stock at Coimbatore works; 24h regional dispatch	6–8 weeks lead time for imported spare parts

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- 100% Helium mass-spectrometer refrigeration circuit leak detection (<0.1 g/year)
- Hydrostatic pressure proof test at 1.5× maximum allowable working pressure (24 bar)
- Continuous 8-hour calorimeter test verifying +3°C PDP at full rated CFM load
- Dielectric high-voltage insulation and ground bond safety testing

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 24 Months comprehensive manufacturer warranty on heat exchanger; 12 months on compressor package. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

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